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# Integrating pattern grammar and local grammar into the identification of constructions and the development of a pedagogical constructicon in EAP context: An exploratory study

Hang Su [a,*], Jun Ye [a], Naixing Wei [b]

[a] Sichuan International Studies University, PR China
[b] Beihang University, PR China

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ABSTRACT

This study explores the possibility of integrating pattern grammar and local grammar into construction grammar research and the value of such research for EAP writing pedagogy. Using data taken from a corpus compiled from Linguistics research articles, our study shows that local grammar analyses of grammar patterns help to identify 'pattern-meaning' combinations. We further describe the functions of such combinations in academic contexts and argue that they can be interpreted as constructions at the mid-level of specificity. We also demonstrate the organisation of these constructions into a constructicon for pedagogical purposes. The implications and applications are subsequently discussed, highlighting in particular that the constructicon not only offers an inventory of pattern-meaning pairings available to express meanings, but also provides quantitative information about the association between verbs and constructions. Overall, our study indicates that the integration of pattern grammar, local grammar and construction grammar could be of immense potential value for EAP writing research and pedagogy.

## 1. Introduction

This paper reports on an exploratory study that explores the possibility of integrating pattern grammar (Hunston & Francis, 2000), local grammar (Hunston, 2002; Hunston & Su, 2019) and construction grammar (Goldberg, 1995, 2006) into the development of a pedagogical constructicon, i.e., a list or network of constructions (Lyngfelt et al., 2018), in EAP context. It aims to show that the local grammar analyses of grammar patterns can facilitate the identification of constructions and the subsequent development of constructicon. It will also discuss the implications and applications of this line of research for EAP writing pedagogy. The study partly aligns with those EAP writing studies which have revealed lists of lexicogrammatical resources such as academic formulas (Simpson-Vlach & Ellis, 2010), academic collocations (Ackermann & Chen, 2013; Lei & Liu, 2018), phrase frames (Cunningham, 2017; Lu et al., 2018) and multi-word units (Rogers et al., 2021). More importantly, the study aligns with existing research that has argued for an applied or pedagogical construction grammar (e.g., Boas, 2022; De Knop & Gilquin, 2016; Holme, 2010) and moves much further to develop a pedagogical constructicon (Patten & Perek, 2022).

The rationale of doing so relates to the basic argument made in construction grammar that the knowledge of a language is the knowledge of constructions (Goldberg, 1995, 2006). Accordingly, constructions should be the focus of language teaching and learning

* Corresponding author. Centre for Foreign Languages and Literature, 33 Zhuangzhi Road, Shapingba District, Chongqing, 400031, PR China.
*E-mail addresses:* suhangunique@hotmail.com (H. Su), shesyvonne@hotmail.com (J. Ye), nxwei@buaa.edu.cn (N. Wei).

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(Ellis et al., 2016, p. 45). It needs to be pointed out, however, that construction grammar has only described in detail a relatively small number of constructions. As Hunston (2019: 324) notes, if construction grammar "is to be made of use by the language pedagogy community, then a way of extending those descriptions to be equivalent to the coverage of more standard grammars of English […] must be found". To that end, Hunston's (2019: 334) proposal is to align pattern grammar with construction grammar, noting that "the pattern-meaning combination can be used as the basis for identifying candidate constructions". For instance, Hunston's (2019) analyses of the pattern **V n *of* n** identified three candidate constructions, as shown in Table 1.

An important notion here is 'pattern-meaning combination', which involves the mapping of meaning elements onto pattern elements, as illustrated in Table 1. We agree that such meaning elements are useful for analysing grammar patterns, which can in turn lead to the establishment of pattern-meaning combination. However, it is unclear how such meaning elements can be proposed or identified; or in Hunston's (2019: 336) own words, "[t]he choice of meaning element labels […] may be controversial". In consequence, one potential problem that needs to be further addressed relates to the identification of meaning elements for analysing grammar patterns. To address this issue, we argue that the currently emerging local grammar research would be useful for identifying such meaning elements and that analysing grammar patterns from a local grammar perspective could help to identify pattern-meaning combinations, i.e., candidate constructions, in a more principled and finer-grained manner. In fact, Hunston (2019) does mention that the identification of semantic elements is influenced by work on local grammar, in addition to others such as frame semantics (Fillmore et al., 2003) and corpus pattern analysis (Hanks, 2013). More notably, Hunston and Su (2019) have already applied local grammars into the analyses of adjective complementation patterns, which shows that such analyses can contribute to identifying evaluative constructions.

Therefore, in this study we draw on and extend Hunston and Su's (2019) and Hunston's (2019) discussion to identify constructions that are context-specific, i.e., constructions that are useful for language teaching and learning in a specific context. Specifically, we take the grammar pattern **V that** (e.g., *argue that, suggest that, note that*) as the starting point and analyse instances of **V that** from the local grammar perspective, aiming to identify constructions and develop a constructicon that are useful for academic writing teaching. Generally speaking, verbs occurring in this pattern are roughly regarded as reporting verbs; hence the reason for focusing on this particular grammar pattern is partly because reporting has been shown to be an important convention and strategy for making an academic text accessible and convincing (Batchelor, 2023; Thompson & Ye, 1991). In particular, Hyland (2002) noted the importance of appropriate uses of reporting verbs in academic writing,

> The selection of a particular reporting verb is a delicate choice as it is a crucial means of both situating one's work appropriately and communicating with one's peers effectively, a way of engaging with colleagues and of appealing to the epistemological and interactive understandings of one's community.
>
> (Hyland, 2002, p. 130)

In addition, research has revealed that **V that** occurs quite frequently in academic discourse (e.g., Charles, 2006a, b). As such, focusing on **V that** could greatly help showcase how the local grammar analyses of grammar patterns can facilitate the identification of constructions and the subsequent development of constructicon. It could also help explore the pedagogical value of the kind of constructicon developed for EAP writing pedagogy.

The remainder of the paper is organised into five further sections. In Section 2 we introduce pattern grammar, construction grammar and local grammar, and discuss their similarities and differences, aiming to provide the theoretical background for the current investigation. Section 3 reviews previous studies that have explored the pedagogical value of either pattern grammar, construction grammar or local grammar in EAP context. Section 4 discusses the data and methodology used in the current investigation. Section 5 presents the local grammar analyses of the **V that** pattern and discusses how the analyses can help to identify constructions and the subsequent development of constructicon. In addition, the implications and applications of the current research will be discussed in detail, highlighting in particular its value for both construction grammar research and EAP writing pedagogy. Section 6 concludes the study and points out directions for future studies.

**Table 1**
**V n *of* n** and constructions (adapted from Hunston, 2019).

| The 'deprive someone of something' construction | | | | | |
|---|---|---|---|---|---|
| Pattern elements | Noun | Verb | Noun | of | Noun |
| Meaning elements | Remover | remove-neg | Possessor | | POSSESSION |
| Example | They | strip | them | of | their identity |
| **The 'inform someone of something' construction** | | | | | |
| Pattern elements | Noun | Verb | Noun | of | Noun |
| Meaning elements | informer | Inform | informed | | message |
| Example | The doctor | had not warned | him | of | the risks of surgery |
| **The 'convict someone of something' construction** | | | | | |
| Pattern elements | Noun | Verb | Noun | of | Noun |
| Meaning elements | attributor | attribute | attributee | | bad action |
| Example | Magistrates | convicted | him | of | breaking a teacher's nose |

## 2. Pattern grammar, construction grammar and local grammar

This section discusses briefly pattern grammar, construction grammar, and local grammar (see also Su & Wei, 2023). Pattern grammar was built on earlier works such as Sinclair (1991), Francis (1993) and Hunston and Francis (1998). It was later on systematically described in Hunston and Francis (2000) and was established as an approach to English language description that prioritises the behaviour of individual words. According to Hunston and Francis,

> The patterns of a word can be defined as all the words and structures which are regularly associated with the word and which contribute to its meaning. A pattern can be identified if a combination of words occurs relatively frequently, if it is dependent on a particular word choice, and if there is a clear meaning associated with it.
>
> (Hunston & Francis, 2000, p. 37)

This indicates clearly the criteria for pattern identification: frequency, co-selection of word choices, and meaning dependency, which provides the basis for excluding those instances which look similar in terms of surface structure but are not real instances of a given pattern (see Section 4 for more discussion).

Two major observations made in pattern grammar are the inseparability between lexis and grammar and the association between pattern and meaning, which are discussed in turn. One example that Hunston and Francis (2000) used to illustrate the first observation is the phrase *want to*. They noticed that *want* often co-occurs with a to-infinitive clause; however, it is unclear "whether this is a fact about lexis (the collocation of *want* and *to*) or a fact about grammar" (ibid: 251). Much evidence has been presented to illustrate the second observation. Most notably, Francis et al. (1996, 1998),[1] have shown that words occurring in each pattern can be divided into a limited number of meaning groups.

Despite the innovation for describing English language offered by, and notable observations (especially the pattern-meaning association) made in, pattern grammar, one of the criticisms that pattern grammar faces is that the proposed meaning groups for each pattern were not guided by any semantic information (Teubert, 2007). Hunston similarly notes that,

> One of the issues in pattern grammar research is the relationship between pattern and semantic classification. Francis et al. (1996, 1998) somewhat naively group words into ad hoc 'meaning groups'. This has led to criticism […] that the groupings are misleading because they appear to propagate the mistaken assumption that meaning 'belongs to' single words and that groups of words can 'mean the same thing', as well as ignoring aspects of semantic and grammatical theory that might more usefully inform the description.
>
> (Hunston, 2011, p. 123)

This points to the need to integrate semantic analysis into the proposal of meaning groups associated with each grammar pattern. We argue that local grammars can offer such semantic analysis that can usefully guide the proposal of meaning groups for each pattern, based on the assumption that words sharing a given local grammar pattern can be grouped into the same meaning group, as will be discussed in greater detail in Section 5.

The second notion that needs to be made clear is construction grammar. On the one hand, pattern grammar and construction grammar are in many ways similar to each other. For example, they both reject the traditional distinction made between lexis and grammar and emphasize the connection between form/pattern and meaning. On the other hand, the two are different, in that pattern grammar is more concerned with language patternings that are observable in actual language use, whereas construction grammar is more concerned with the mental representation of such patternings (see also Hunston, 2014).

The theoretical origin of construction grammar can be traced back to the work of Fillmore, in particular case grammar (Fillmore, 1968) and frame semantics (Fillmore et al., 2003), as well as Langacker's (1987, 1991) cognitive grammar (see more discussion in Ungerer & Hartmann, 2023). It must also be pointed out that there are many approaches to construction grammar (see Hoffmann & Trousdale [2013] for more discussion) and the approach that we draw on is that of Goldberg (1995, 2006). Simply put, constructions are form-meaning pairings (Goldberg, 1995), defining criteria of which are further elaborated:

> Any linguistic pattern is recognized as a construction as long as some aspects of its form or function is not strictly predictable from its component parts or from other constructions recognized to exist. In addition, patterns are stored as constructions even if they are fully predictable as long as they occur with sufficient frequency.
>
> (Goldberg, 2006, p. 5)

Like pattern grammar, the definition of construction also points to the criteria for identifying constructions, i.e., form-meaning pairing, unpredictability of meaning, and frequency of occurrence. In particular, Goldberg's (2006) observation that frequently occurring patterns can also be stored as constructions lends support to our proposal of taking grammar patterns as the starting point to identify constructions. As will be shown below, we rely on these criteria for identifying constructions.

Constructions can occur at different levels of complexity, ranging from morphemes and words to phrases and abstract syntactic patterns, and even to texts or genres; or in Goldberg's (2006: 18) words, "it's constructions all the way down". Constructions that have been examined include those specific ones such as 'accident waiting to happen' as well as those general ones such as the 'ditransitive'

[1] The two volumes of Grammar Pattern series are now out of print, but the publisher has made it publicly available online via https://grammar.collinsdictionary.com/grammar-pattern.

construction (e.g., Goldberg, 2006). Hunston and Su (2019) commented on this multi-level approach of construction grammar, noting that this is both an advantage and disadvantage. The advantage is that "all of lexis and grammar can be described in a single model, without the need for an elaborate system of grammatical levels or ranks", whereas the drawback is that "the number of potential constructions is vast, and a listing of them all seems an impossible task" (ibid: 570). In other words, this constructions-include-all approach inevitably raises the question of how to systematically identify constructions. This paper will address this issue and argue that the kind of pattern-meaning combination offered by local grammar analysis can help us to identify constructions at a mid-level of specificity, i.e., those constructions which are neither too general (e.g., ditransitive construction) nor too specific (e.g., constructions at the morpheme or word level).

In construction grammar, "language is seen as an inventory of constructions, defined as form-meaning/function pairings, and language learning is in essence a piecemeal, frequency-biased process of accumulating these constructions" (Liu & Lu, 2020, p. 1). If language is seen as an inventory of constructions, then identifying constructions and developing pedagogical constructicon would be essential for language teaching and learning. It is then not surprising that much effort has been made to develop constructicons (e.g., Lyngfelt et al., 2018). In the enterprise of developing constructicon in English, researchers have either relied upon FrameNet (Fillmore et al., 2012) or combined frames with grammar patterns (Perek & Patten, 2019). In the present study we propose an alternative approach to identify constructions and apply it to develop constructicon in academic context. Our proposed approach starts from the selection of grammar patterns, then the local grammar analyses of the selected pattern, subsequently the re-interpretation of pattern-meaning combination as constructions, and finally to organise the identified constructions into a constructicon, as will be demonstrated in Section 5.

The third notion that the present study draws upon is local grammar. Briefly, local grammar is an alternative approach, as opposed to traditional or general grammars, to describing language use that seeks to account for one meaning or function only (Hunston, 2002). In other words, unlike general grammars which seek to generalise rules that govern language use in various contexts, each local grammar aims to describe a subset of language use that is associated with a specific meaning or function. The most notable difference between the two lies in the analysis. Specifically, general grammars use terms such as Subject, Predicate and Object to linguistic analysis and description, whereas local grammars use context-specific functional terms to analyse language use associated with a specific meaning or function. Compared with general grammar analysis, the advantage of local grammar analysis pertains to the revelation of the discourse function of the utterance under examination. For instance.

(1) I'm sorry that our time is up. (KRL 4562)
(2) I'm pleased that that's been clarified. (JAC 417)

The general grammar analysis and the alternative local grammar analysis of the two examples are presented in Table 2.

There are two points worth noting here. First, although Table 2(a) is only a simplistic rendering of general grammar analysis, it shows that general grammar offers a way to describe language uses that may have different functions or meanings; such that the 'Subject – Predicate – Complement' sequence can not only be used to describe expressions that are related to emotion-reporting, but also to expressions that are related to apology-making. The shortcoming, however, of the general grammar analysis is that such an analysis cannot reveal the distinction between the discourse function of expressions that look structurally similar. In contrast, the kind of analysis offered by local grammars can more easily reveal the discourse function of the two instances, although it is more restricted or specialised. Second, the kind of local grammar analysis leads to the establishment of pattern-meaning combination, or local grammar patterns, which connects closely form with function. As construction is primarily defined as form-meaning/function pairing, it can thus be argued that this kind of pattern-meaning combination can be recognised as constructions (see also Hunston, 2019; Hunston & Su, 2019). This is the main argument of the present study, and in Section 5 we demonstrate how local grammar analyses of grammar patterns can help to establish pattern-meaning combinations, which in turn leads to the identification of constructions and the development of constructicon.

In addition, it is worth mentioning that local grammar analysis is consistent with the thematic role analysis used in construction grammar.[2] For instance, construction grammar uses thematic roles such as Agent, Recipient, Beneficiary etc. to analyse argument structure constructions (Goldberg, 1995, 2006). The difference between the kind of functional analysis in local grammar and that in construction grammar lies in the functional terms used in each. Basically, local grammar uses functionally more transparent but less general, context-specific terminologies, whereas construction grammar uses more general but functionally less transparent ones. Nevertheless, the fact that both involve functional analysis indicates the compatibility between local grammar and construction grammar research.

In this section we have briefly introduced the three approaches and discussed the similarities and differences between them, providing the theoretical background for the subsequent discussion. To recapitulate, pattern grammar and construction grammar share the viewpoints that lexis and grammar are inseparable and that form/pattern and meaning are associated, whereas they differ from each other with regard to the kind of patternings each focuses on, i.e., pattern grammar focuses on patternings that are observable in

[2] Equally importantly, it merits mentioning that local grammar has much in common with Systemic Functional Grammar (SFG; Halliday & Matthiessen, 2004), especially that the mapping of functional labels onto corresponding formal elements in local grammar analysis is similar to that of transitivity analysis in SFG. There are also differences between the two, however. Specifically, the labelling of semantic roles in transitivity analysis depends upon the process type a clause belongs to, whereas the local grammar labelling relies on the discourse function of a clause (see more discussion in Hunston & Francis, 2000, pp. 128–129).

**Table 2(a)**
The general grammar analysis.

| Subject | Predicate | Complement |
|---|---|---|
| *I* | *'m sorry* | *that our time is up* |
| *I* | *'m pleased* | *that's been clarified* |

**Table 2(b)**
The alternative local grammar analysis.

| Apologiser | Hinge | Apologising | Specification |
|---|---|---|---|
| *I* | *'m* | *sorry* | *that I have to turn down the invitation* |
| Emoter[#] | Hinge | Emotion | Trigger |
| *I* | *'m* | *pleased* | *that I was invited to give a talk at …* |

[#] The terms are taken from Bednarek (2008).

actual language use whereas construction grammar on the mental representation of such patternings. Local grammar and construction grammar are similar in that each involves the use of functional terms to analyse language use, though differing in the degree of transparency and generalisability of the functional terms used in each.

## 3. Pattern grammar, construction grammar and local grammar in EAP context

There have already been a few studies which explored the value of either pattern grammar, construction grammar, or local grammar in EAP writing research and pedagogy. For example, Green (2019) offered a systematic description of the grammatical patterns of those academic vocabularies identified in the Academic Word List (Coxhead, 2000) and Secondary Vocabulary List (Green & Lambert, 2018), and further translated the grammatical description into a pedagogical resource for vocabulary teaching and learning. In another study, Ma and Qian (2020) proposed a method for extracting grammar patterns and applied it to create a list of grammar patterns for verbs that frequently occur in academic writing. Liu and Lu (2020) moved even further. They presented an attempt that combined pattern grammar and construction grammar to the analyses of *N1 of N2* construction in disciplinary academic writing. Their analyses lead to the identification of five functional categories, including Event, Support, Focus, Proper nouns and Quantification/Classification. What is particularly worth noting is that their study offered "support for the effectiveness of the pattern grammar approach to phraseology and to the development of pedagogically useful constructions of academic English" (Liu & Lu, 2020, p. 10). In the present study we further explore how pattern grammar can contribute to identifying constructions in a more systematic and principled manner.

In addition to those studies which explored the pedagogical application of pattern grammar and construction grammar, there have also been few studies which have either attempted to assess second language (L2) writing quality by examining constructional complexity or to develop profiles of English verb argument constructions (VACs) in academic contexts. For instance, Mostafa and Crossley (2020) used VAC-based indices to measure writing quality and found that the relationship between VAC-based predictors and L2 writing quality is largely task-dependent. Casal et al. (2022) examined genre and discipline effects on the use of 11 VACs in research articles across four disciplines. Their investigation revealed that verbs occurring in each VAC differ considerably across general and specialised corpora and that there is a strong disciplinary effect on the use of VACs across disciplines, such as the frequency of VACs vary across disciplines, and VACs occur more frequently in social science RAs. Based on the pervasiveness and productivity of VACs, Casal et al. argue that VACs have significant pedagogical value and the VAC list provided in Francis et al. (1996) is highly beneficial.

As discussed in Section 2, the present study also draws on insights from local grammar which has been increasingly applied in EAP writing research. Existing studies that used local grammar in academic discourse studies mainly focused on the description of discourse acts, i.e., rhetorical function such as giving examples, defining terms, making comparison and contrast, etc. For instance, Su and Zhang (2020) used the local grammar approach to describe how academic writers give examples to illustrate or support an argument/proposition or to specify abstract superordinate categories. Similarly, Zhang and Su (2022) employed the local grammar approach to describe the linguistic realisations of definition in Linguistics research articles. Another line of research that applied local grammar to academic discourse studies focused on investigating the similarities and differences between EFL learners' and expert writers' use of discourse acts in academic writing. For instance, Su et al. (2022) explored the ways in which Chinese EFL learners and expert writers provide examples and found that, although the two groups of writers used exemplification at similar frequency, Chinese EFL learners appear to be not as familiar as expert writers with the strategy of exemplifying by citing relevant studies. In contrast, Zhang and Su's (2023) investigation into Chinese EFL learners' and expert writers' use of definition revealed a different picture, that Chinese EFL learners performed definition much more frequently than expert writers did, which is likely due to that EFL learners may consider definition "a general writing strategy – a 'filler' strategy – to meet the size requirements of the writing task" (Ädel, 2006, p. 110).

What these studies have shown is that, first, local grammars are useful for documenting and describing the linguistic realisations of a given discourse act, and the resulting local grammars make available to EFL learners an inventory of the lexicogrammatical realisations associated with that discourse act, thereby enriching EFL learners' repertoire to perform that discourse act. Second, such investigation offers important insights into the phraseological tendency of language use, in particular that it shifts our understanding of

phraseology from lexicogrammatical patterning towards discourse-semantic patterning. Third, the investigation into EFL learners' and expert writers' use of discourse acts allows one to diagnose what problems or difficulties EFL learners may have in performing such discourse acts and therefore to further offer pedagogical implications for academic writing teaching (especially in EFL contexts).

Although studies have discussed the respective pedagogical value of pattern grammar, construction grammar, and local grammar in EAP context, few attempts have yet been made to adequately explore their applications for EAP writing teaching. In the case of construction grammar, Casal et al. (2022) similarly note that, although constructions, especially VACs, have been the focus of recent research adopting a usage-based approach to language teaching and learning, limited attention has been paid to VACs in academic writing pedagogy. Furthermore, it also appears that there have been even fewer studies that explored how patterns and constructions can be incorporated into EAP teaching and measured the teaching outcomes, indicating that more classroom intervention is needed. Therefore, in the present study we combine the three approaches and explore their value in EAP writing research and pedagogy. We take the **V that** pattern as the starting point and showcase how local grammar analyses of the pattern can facilitate the identification of constructions and the subsequent development of constructicon. The pedagogical implications and applications of this kind of research in EAP writing pedagogy will also be discussed.

## 4. Corpus and data retrieval

For the current investigation, we complied a corpus by collecting Linguistics research articles published between 2010 and 2019 in ten leading international journals. These journals cover a wide range of subfields of Linguistics, such as applied linguistics, semantics, pragmatics, cognitive linguistics etc. (see Table 3). Briefly, we decided to select Linguistics research articles as the academic genre of focus because they primarily function to construct and disseminate new knowledge or findings (Bazerman, 1988). This means that academic writers need to frequently use the reporting pattern **V that** to either report observations or findings made in existing studies or those drawn in their own research, which leads to Hyland's (2002) argument that reporting is an important convention of academic writing. Indeed, other studies such as Charles (2006a) also found that reporting clauses are considerably more frequent in social science disciplines than in disciplines of natural sciences, and Gray (2015) observed higher use of verb complement clauses in humanities and social sciences but lower use in hard sciences. Furthermore, we selected research articles published in these ten leading journals because of their high impact in their respective subfields. These properties ensure that research articles sampled in the study are highly representative of academic writing in the field of Linguistics. At this point, it should be mentioned that the L1 backgrounds of the authors of these sampled articles were not considered in the process of data collection, because all published articles in these top-ranking journals have been rigorously peer reviewed and carefully edited after acceptance, and as such language use in these articles is acceptable to the academic community. The metainformation of the corpus is given in Table 3.

We then uploaded the corpus to Sketch Engine (Kilgarriff et al., 2004) for data processing. We searched for the **V that** pattern[3] and, for the sake of manageability, analysed those verbs which occurred more than ten times in our corpus. This process left us 57 verbs, amounting to 4343 instances. Table 4 provides the basic information of these verbs.

Three issues concerning data retrieval need to be further clarified. The first relates to the decision to omit cases of *that* omission. This is partly because research has shown that retention of *that* is the norm in academic writing (Charles, 2006a; Hyland & Tse, 2005), although omission of *that* complementiser is possible. Additionally, this is also due to technical problems, as it is quite challenging for computerised search to exhaustively identify instances of **V that** (especially in a relatively large corpus such as the one used in the present study) if the *that*-indicator is absent. In our future research we will explore if parsers like the Stanford NLP or SPACY could help overcome this problem.

The second issue is concerned with the identification of instances genuinely realises the **V that** pattern. In Section 2 we discussed the criteria for pattern identification: frequency, co-selection of word choices, and meaning dependency (Hunston & Francis, 2000). In other words, the identification of a pattern does not only rely upon the frequency of the co-occurrence between a verb and the *that*-clause, but more importantly the meaning has to be dependent upon the co-occurrence between the verb and the *that*-clause. Indeed, a manual scan of our data confirms that although some verbs frequently co-occur with a *that*-clause, they do not qualify as instances of the **V that** pattern. Take 'GIVE that' as an example. In the instance *This is of particular interest given that the signers come from backgrounds that use different scripts*, the verb *given* was not used to project an idea or argument and *that* was not used to introduce a complement clause. This is also true for other cases such as 'SEEM that' and 'PROVIDE that', which are not considered as instances instantiating the **V that** pattern in the present study.

The final issue has to do with the pattern ***it be* V-ed that**. Basically, this pattern means that the verb (V) is preceded by an introductory *it* and a *be* verb in various tenses, and is followed by a *that*-clause, as in *it has been suggested that mental states of surprise arise from a contrast between the expectation* … While this pattern has been analysed as the passive form of **V that** in studies such as Charles (2006a, 2006b), we strictly followed Francis et al.'s (1996: 526–528) description and considered it as a distinct pattern. In other words, instances of ***it be* V-ed that** were excluded out of the present study.

## 5. Analyses, results, and discussion

In this section we present the local grammar analyses of the **V that** pattern and discuss how the local grammar analyses can

[3] The query script for **V that** in Sketch Engine is '[tag = "V.*"] [tag = "IN/that"]'.

**Table 3**
Metadata of the corpus used in the present study.

| Journals | No. of texts | No. of tokens | Average length |
|---|---|---|---|
| *Applied Linguistics* | 10 | 75,439 | 7543.9 |
| *Cognitive Linguistics* | 10 | 101,866 | 10,186.6 |
| *Journal of Linguistics* | 10 | 118,697 | 11,869.7 |
| *Journal of Pragmatics* | 10 | 76,653 | 7665.3 |
| *Journal of Semantics* | 10 | 126,400 | 12,640 |
| *Language* | 10 | 157,804 | 15,780.4 |
| *Language in Society* | 10 | 87,898 | 8789.8 |
| *Language Teaching* | 10 | 152,909 | 15,290.9 |
| *Modern Language Journal* | 10 | 97,933 | 9793.3 |
| *Research on Language and Social Interaction* | 10 | 83,600 | 8360 |
| **Total** | **100** | **1,079,199** | **10,791.99** |

**Table 4**
The top 57 verbs co-occurring with **V that** in our corpus.

| Lemma | Frequency | Lemma | Frequency | Lemma | Frequency |
|---|---|---|---|---|---|
| SUGGEST | 591 | OBSERVE | 63 | HOPE | 18 |
| SHOW | 456 | BELIEVE | 57 | EXPECT | 17 |
| ARGUE | 380 | ENSURE | 48 | SUPPOSE | 17 |
| FIND | 321 | REPORT | 45 | AGREE | 16 |
| ASSUME | 190 | IMPLY | 44 | HYPOTHESISE | 15 |
| NOTE | 174 | SIGNAL | 40 | HOLD | 15 |
| INDICATE | 167 | APPEAR | 36 | ALLOW | 15 |
| MEAN | 154 | SEEM | 35 | FEEL | 14 |
| PROPOSE | 127 | CONFIRM | 33 | PROVIDE | 14 |
| DEMONSTRATE | 119 | PRESUPPOSE | 32 | LEARN | 14 |
| SAY | 107 | STATE | 31 | EXPLAIN | 13 |
| CONCLUDE | 92 | ASSERT | 30 | TELL | 12 |
| KNOW | 86 | FOLLOW | 29 | DISCOVER | 11 |
| PREDICT | 86 | REQUIRE | 28 | POSIT | 11 |
| CLAIM | 80 | ACKNOWLEDGE | 28 | REALISE | 11 |
| THINK | 79 | ENTAIL | 25 | MAINTAIN | 11 |
| REVEAL | 76 | ESTABLISH | 23 | CONSIDER | 10 |
| GIVE | 70 | MENTION | 21 | UNDERSTAND | 10 |
| SEE | 67 | RECOGNISE | 19 | NOTICE | 10 |

contribute to the identification of constructions and the subsequent development of constructicon. The theoretical and pedagogical implications will also be discussed in detail.

### 5.1. Local grammar analyses of V that

As discussed in Section 2, the peculiarity of local grammar research is that it uses context-specific and functionally transparent terms to analyse the corresponding syntactic units. It must be noted that such terms were first identified based on preliminary sample analyses, and further revised throughout the whole process of data analyses. Nevertheless, when we initially attempted to perform the local grammar analyses of **V that**, we drew on insights from the terms used in Charles's (2006a, 2006b) analyses of reporting clauses and frame elements in FrameNet (https://framenet.icsi.berkeley.edu/about) (although we also observed that in most cases the kind of frame elements used in frame semantics does not apply to our local grammar analyses of those verbs co-occurring with the **V that** pattern in the genre of academic writing).

With regard to the analyses, we divided instances of the **V that** pattern into three components: the nominal group on the left of the node pattern, the verb, and the *that*-clause. For the nominal group occurring on the left of the node pattern, we identified six types of functional categories, three of which indicate the source of a given information (e.g., argument, statement). Specifically, based on Charles' (2006b) observation that reporting clauses can be used to either report other researchers' work or the writer's own work, we made the distinction between 'Source-external' and 'Source-internal', as illustrated in examples (3)–(5).

(3) Scheurich (1995: 241) argues that we need to take more account of 'complexity, uniqueness, and the indeterminateness' of …
(4) This article argues that this is achieved through interviewer turns that are predominantly assessment oriented.
(5) We argue that such meanings need to be taken seriously.

In Example 3, the writer attributes the argument to 'Scheurich', whereas in Examples (4) and (5), the argument was attributed to 'This article' and 'We', i.e., the writer him/herself, which were thus labelled 'Source-external' and 'Source-internal' respectively. This

also means that 'Source-external' was roughly indicated by the presence of citation and that 'Source-internal' was normally realised by wordings such as *I, we, this article, this study, the current research* etc.

In addition to the above two types of sources, we also observe a third type, namely, research-participant. For example.

(6) When interviewed, the teachers in Duff & Polio's study (1990) suggested that they were more likely to use the learners' own language …

In Example (6), the source of the suggestion is *the teachers* being interviewed, who served as the research participants involved in *Duff & Polio's study*. Hence, we analysed *the teachers* as 'Research-participant'.

In addition to these animate subjects, the nominal group occurring on the left of the node pattern **V that** may also refer to inanimate ones, such as *table, data, analysis, requirement, criteria* etc, as illustrated in examples (7)–(9).

(7) The data shows that speakers switch to SA for eight main reasons.
(8) These results suggest that pronunciation during interaction warrants more investigation.
(9) The linguistic selection criteria ensured that the four participants had no language in common at the beginning of the research period.

In example (7), *the data* is a kind of evidence that causes the writer to make interpretation. We thus labelled it as 'Evidence', which was typically realised by research nouns (e.g., *the data, the analysis*) and textual nouns (e.g., *the table, the figure*). In contrast, in Example (8) the writer is not making any interpretation about evidence, but about results, which was thus labelled 'Finding'. Example (9) is different in that the subject *The linguistic selection criteria* is kind of something that makes sure something else happen or causes something else to happen, which was thus labelled 'Causer'.

In above we have discussed the semantics of the subject of instances instantiating the **V that** pattern. The remaining is the semantics of the verb and the *that*-clause, which is relatively simpler to analyse. Our local grammar analyses suggest that these verbs can be grouped into seven semantic categories (cf. Charles, 2006a, 2006b; Francis et al., 1996): ARGUE, FIND, CAUSE-TO-INTERPRET, CAUSE-TO-HAPPEN, THINK, HYPOTHESISE and STATE and, correspondingly, the *that*-clause into ARGUMENT, FINDING, INTERPRETATION, EFFECT, THOUGHT, HYPOTHESIS and STATEMENT. Based on the above discussion, we can summarise the analytic framework for a local grammar analysis of **V that** in Fig. 1.

Using the analytic framework, we then performed the local grammar analyses of the **V that** pattern and presented the results in Table 5.

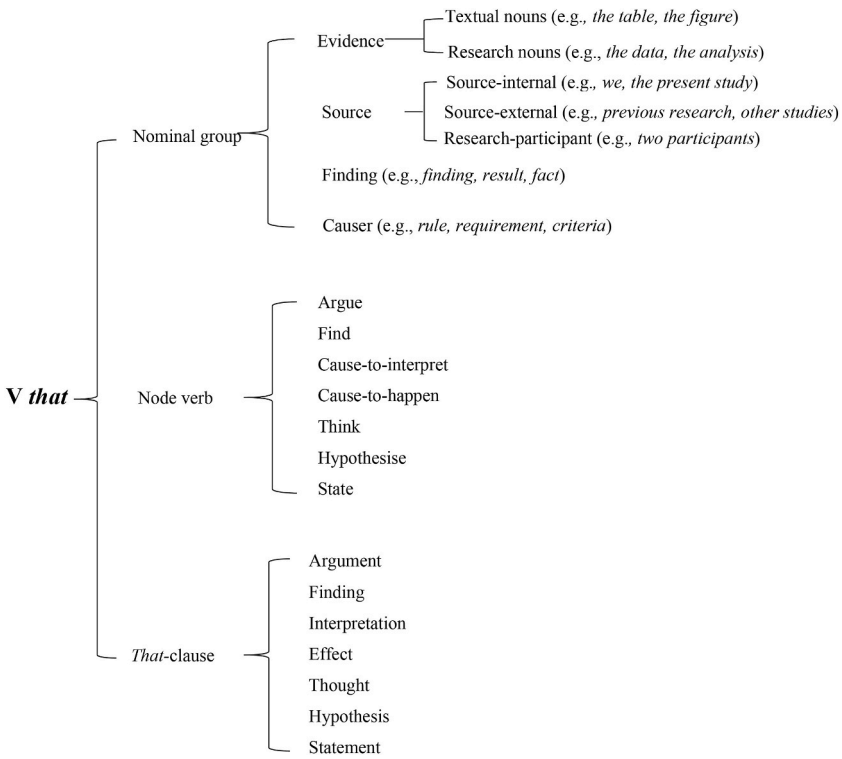


**Fig. 1.** The analytic framework for a local grammar analysis of **V that**.

**Table 5**
An overview of local grammar analyses of **V that**.

| **PA 1** | **Evidence** | **Cause-to-interpret** | **that** | **Interpretation** |
|---|---|---|---|---|
| 714[5] | These data | suggest | that | the active filler strategy is sensitive to … |
| **PA 2** | **Source-external** | **Find** | **that** | **Finding** |
| 634 | Large-scale empirical studies | have found | that | some of the predicted effects are indeed significant |
| **PA 3** | **Source-external** | **Argue** | **that** | **Argument** |
| 597 | Schegloff (2001) | argued | that | the interlocutors use the word *no* to … |
| **PA 4** | **Source-internal** | **Find** | **that** | **Finding** |
| 378 | We | find | that | subjects rely on alternation-based generalizations … |
| **PA 5** | **Source-internal** | **Argue** | **that** | **Argument** |
| 295 | We | argue | that | the gap restriction can be maintained … |
| **PA 6** | **Finding** | **Cause-to-interpret** | **that** | **Interpretation** |
| 288 | The results | showed | that | L2 speakers were able to recognize different interactional settings … |
| **PA 7** | **Research-participant** | **State** | **that** | **Statement** |
| 152 | In interview Mandeep | had said | that | if you come from Punjab to Southall ... |
| **PA 8** | **Research-participant** | **Think** | **that** | **Thought** |
| 148 | The NS teachers | thought | that | major obstacles for team-teaching included … |
| **PA 9** | **Source-internal** | **Hypothesise** | **that** | **Hypothesis** |
| 118 | ... we | hypothesised | that | there would be differences in temporal reasoning … |
| **PA 10** | **Source-internal** | **Think** | **that** | **Thought** |
| 59 | We | believe | that | this proposal paves the way for … |
| **PA 11** | **Causer** | **Cause-to-happen** | **that** | **Effect** |
| 43 | … formal coding | requires | that | social behaviour be reduced to ... |
| **PA 12** | **Source-external** | **Hypothesise** | **that** | **Hypothesis** |
| 31 | Hauser et al. (2009) | hypothesized | that | anger and the Moving Ego representation of time may tap into … |
| **PA 13** | **Research-participant** | **Find** | **that** | **Finding** |
| 23 | ... parents with limited L2 proficiency | concluded | that | non-involvement was the best strategy ... |
| **PA 14** | **Source-external** | **Think** | **that** | **Thought** |
| 16 | Pesetsky (1995) | thought | that | command phenomena and constituency tests pointed in conflicting directions ... |
| **PA 15** | **Source-external** | **State** | **that** | **Statement** |
| 12 | Cazden (1992) | explains | that | for these U.S. “structured English immersion” programs, the term “immersion” was ... |
| **PA 16** | **Source-internal** | **State** | **that** | **Statement** |
| 10 | When we | say | that | the MIP is satisfied in a certain situation ... |
| **PA 17** | **Evidence** | **Source-internal** | **Cause-to-interpret** | **that** | **Interpretation** |
| 4 | By ‘discovery’ | we | mean | that | the speaker is signalling a sharp change ... |
| **PA 18** | **Evidence** | **Source-external** | **Cause-to-interpret** | **that** | **Interpretation** |
| 1 | ... by which | he | means | that | multilinguals are able to communicate with ... |

[5]PA here is the abbreviation of ‘pattern’, and the number in each cell indicates the frequency of each corresponding pattern.

### 5.2. Patterns, local grammars, constructions, and constructicon

As shown in Table 5, the analyses led to 18 local grammars patterns. These local grammar patterns bring together form (i.e., the **V that** pattern), meaning elements (e.g., EVIDENCE, CAUSE-TO-INTERPRET, INTERPRETATION) and function (e.g., reporting finding, interpreting evidence), representing form-meaning-function pairings. Our argument here is thus that the kind of form-meaning-function combination or, more simply pattern-meaning combination, offered by local grammars can be recognised as candidate constructions (see also Hunston, 2019; Hunston & Su, 2019). Specifically, patterns 1, 17 and 18 can be interpreted as 'Cause-to-interpret' construction, in which research evidence was presented and causes one to interpret it. Patterns 2, 4 and 13 can be interpreted as 'Finding-reporting' construction, which was used to either report the writer's own finding or findings drawn by other research(ers). Patterns 3 and 5 can be interpreted as 'Argument-making' construction, in which the writer either report his/her own argument or the argument made in other studies. The remaining patterns can be similarly interpreted: Pattern 6 as 'Finding-interpreting' construction; patterns 7, 15 and 16 as 'Statement-making' construction; patterns 8, 10 and 14 as 'Mental-state-recounting' construction; patterns 9 and 12 as 'Hypothesis-making' construction; and pattern 12 as 'Cause-to-happen' construction.

As our aim is not only to identify constructions, but also to develop constructicon, we follow Hunston and Su's (2019) and Hunston's (2019) discussion and propose a way to describe these constructions so that they can be organised as a network of constructicon. Take the 'Cause-to-interpret' construction as an example. One form associated with this construction is the grammar pattern **V that**; its function is to report how one interprets some sort of research evidence; and at the discourse-semantic level, its meaning elements include 'EVIDENCE', 'CAUSE-TO-INTERPRET', 'INTERPRETATION' and 'SOURCE'. Our corpus analysis also allows us to provide quantitative information, i.e., frequency, for each verb occurring in this construction. Bringing together the above discussion, the constructicon description of this construction is illustrated in Table 6.

To further illustrate, in Table 7 we present the constructicon description of the 'Argument-making' construction.

Because of space constraints, we cannot present the constructicon description for all constructions identified in the present study. Nevertheless, each construction can be described in the same way as shown in Tables 6 and 7 Drawing on the concept of hierarchy of constructicon (e.g., Wible & Tsao, 2020), we can further group these constructions into three broad categories, i.e., communication (communicating information), action (indicating acts or actions) and cognition (reporting mental state). Fig. 2 shows the construction network, i.e., the constructicon, developed in the present study.

### 5.3. Implications and applications

Fig. 2 presents the system network of the **V that** pattern, its local grammar patterns and the constructions identified. It also provides lists of verbs and their frequency of occurrence in each construction. These would have important theoretical implications for construction research and pedagogical applications for EAP writing pedagogy. The theoretical implications relate to the identification of constructions and development of constructicon. As mentioned earlier, construction grammar holds that language is not a system of grammatical rules, but a network of constructions at various levels of schematicity/abstraction and of varying size and complexity. However, as constructions range from morpheme and word to structures and discourses or even genres, current research on construction grammar has not adequately addressed the question as to how constructions can be comprehensively identified. While the study reported in this paper cannot entirely solve this problem either, it offers one possible solution to identify one particular type of construction, i.e., 'mid-level' constructions, by which Hunston and Su (2019) refer to those constructions which are neither too general nor too specific, as discussed in Section 1. Examples of mid-level construction include Goldberg's (2006) 'caused-motion' construction (e.g., *she sneezed the napkin off the table*), 'causative-into' construction (e.g., *they trapped him into a confession*), or the 'Argument-making' construction identified in the present study. Our study shows that local grammar analyses of grammar patterns help to identify pattern-meaning combination that can be subsequently interpreted as constructions, thereby effectively facilitating the identification of such mid-level constructions in a systematic and principled manner.

Furthermore, the analyses show that local grammar research involves more than equating a syntactic structure with a semantic

**Table 6**
A constructicon entry for the 'Cause-to-interpret' construction.

| The 'Cause-to-interpret' construction | | | | |
|---|---|---|---|---|
| Pattern | V that | | | |
| **Function** | Presenting the writer's or other researcher's interpretation of research evidence | | | |
| **Meaning elements** | EVIDENCE, CAUSE-TO-INTERPRET, INTERPRETATION, SOURCE-EXTERNAL, SOURCE-INTERNAL | | | |
| **Pattern-meaning combination** | Nominal group<br>EVIDENCE | Verb<br>CAUSE-TO-INTERPRET | that<br>that | That-clause<br>INTERPRETATION |
| **Example** | Evidence from conversation | suggests | that | visual and other sensory experience is also construed as emblematic of attention |
| **Verbs** | suggest$_{(167)}$[a], show$_{(111)}$, mean$_{(81)}$, indicate$_{(78)}$, predict$_{(44)}$, demonstrate$_{(34)}$, presuppose$_{(28)}$, imply$_{(26)}$, signal$_{(23)}$, entail$_{(21)}$, confirm$_{(17)}$, reveal$_{(13)}$, assert$_{(12)}$, state$_{(11)}$, say$_{(10)}$, assume$_{(10)}$, hold$_{(7)}$, claim$_{(7)}$, posit$_{(4)}$ … | | | |

[a] The number in parentheses indicates the frequency of the verb co-occurring with **V that**, which applies throughout the paper. It is also worth noting that the selection of a specific reporting verb allows one to comment on, or convey his/her attitude towards, the argument, observation or finding being reported (see also Charles, 2006a, 2006b; Thompson & Ye, 1991).

**Table 7**
A constructicon entry for the 'Argument-making' construction.

| The 'Argument-making' construction | | | | |
|---|---|---|---|---|
| Pattern | V that | | | |
| **Function** | Reporting an argument either made by the writer him/herself or by other researcher(s) | | | |
| **Meaning elements** | SOURCE-EXTERNAL, SOURCE-INTERNAL, ARGUE, ARGUMENT | | | |
| **Pattern-meaning combination** | Nominal group | Verb | that | That-clause |
| | SOURCE-EXTERNAL | ARGUE | that | ARGUMENT |
| **Example** | Baker (1989) | Argues | that | serial verb languages have doubly-headed VPs |
| **Verbs** | argue(242), suggest(164), propose(54), note(34), claim(29), indicate(29), acknowledge(10), maintain(9), assert(8), state(7), hold(5), agree(2), posit(2), say(2) … | | | |

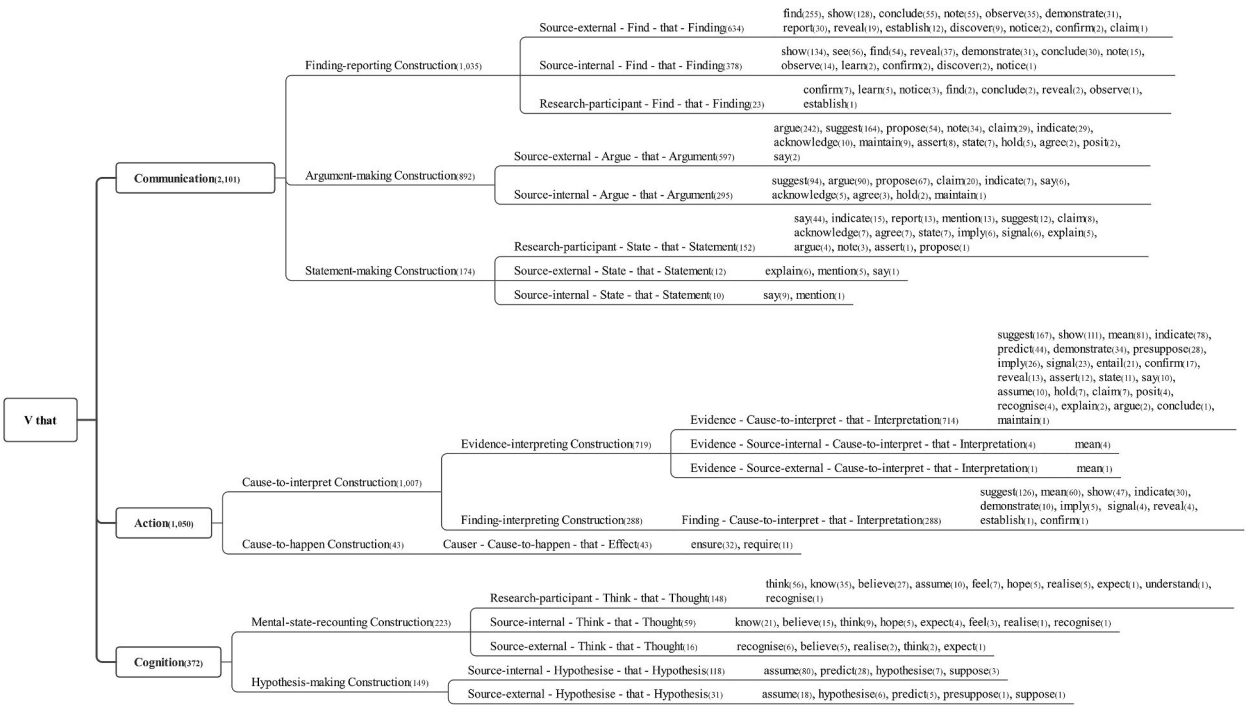


**Fig. 2. V that**, local grammar patterns and the construction network.

pattern; rather, it helps to reveal the different functions or readings a given grammatical structure can have (see also Teubert, 2007). In other words, local grammar analysis allows one to distinguish expressions that look structurally alike but have different semantic readings. In the case of **V that**, for example, the local grammar analyses allow us to identify 18 patterns at the discourse-semantic level, which were in turn interpreted as seven constructions. Based on their semantic association, these constructions are further organised into a network, showing how constructions identified by analysing grammar patterns from a local grammar perspective can contribute to the development of constructicon. It is in these respects that we argue that combining pattern grammar and local grammar can be useful for construction grammar research.

More importantly, the constructions identified and the constructicon developed could have useful pedagogical applications for EAP writing pedagogy (see also Hunston, 2019; Hunston & Su, 2019). In fact, existing studies have discussed comprehensively the pedagogical value of pattern grammar, construction grammar, and local grammar. For instance, Hunston and Francis (2000) argued that pattern grammar can be a useful resource to enrich vocabulary learning and increase accuracy and fluency of language production. Recent research in construction grammar has similarly emphasised the applied or pedagogical value of constructions (e.g., Boas, 2022; De Knop & Gilquin, 2016). For instance, Holme (2010: 117) argues that

> "our notion of the construction should shape both the 'what' and the 'how' of teaching. The 'what' consists of the language content that our notion of a construction identified. It also means deciding whether to treat it as a productive grammatical pattern or a closely specified lexical form. The 'how' originates in the principle of the embodied cognition from which the form and meaning of the symbolic complex have been fashioned and through which they will be learnt".
>
> (Holme, 2010: 117)

The same applies to local grammar research. In particular, Hunston and Su (2019) and Hunston (2019) noted the possibility of

using local grammar patterns to develop a thesaurus-like constructicon, and Su (2020) discussed how findings of local grammar research can be used to develop pedagogical materials for English language teaching.

The present study has further explored the pedagogical applications of this line of research. We argue that the kind of constructicon developed in the present study can be applied into academic writing teaching at least in the following ways. First, it offers an inventory of form-meaning-function pairings, i.e., a constructicon, available to express meanings or to perform functions. For instance, the local grammar analyses suggest that **V that** can be associated with seven functions, with each has its own discourse-semantic patterns and variations. This provides much richer information for the patterns or phraseologies learners may encounter, because "learners need to know what functions they are likely to need to express, what forms most appropriately fulfil those functions, and what variation those forms permit when they are fitted into specific contexts" (Durrant & Mathews-Aydınlı, 2011, p. 61). Moreover, the construction network also offers frequency information about each verb when used to express a specific meaning, as shown in Fig. 2. In other words, the constructicon can help to enrich students' lexicogrammatical choices such as verbs and constructions, in particular highlighting "which meanings are most typically construed by which construction and, what the most common lexical items are in each construction" (Ellis et al., 2016, p. 273).

Following the above discussion, the second pedagogical application of the constructicon developed relates to academic vocabulary teaching and learning. In fact, Green (2019: 2) already noted that "[p]attern grammar and construction grammar, in particular the notion of mid-level constructions, provide a meaningful framework from which to study academic vocabulary". Unlike Green (2019) who focused primarily on describing the grammatical patterns associated with academic vocabularies, the present study has brought this further by, on the one hand, providing the frequency of a given verb in a given construction, thereby raising learners' awareness of the association between verbs and constructions. On the other hand, the study offers pairings of verbs and local grammar patterns/constructions, thereby adding a discourse-semantic dimension to the uses of academic vocabularies. We argue that these can deepen learners' understanding of target vocabularies and therefore substantially contribute to learning academic vocabularies.

Third, the constructicon developed can be usefully incorporated into the design and development of academic writing teaching materials. Herbst (2016) proposed several principles for pedagogical construction grammar. Particularly relevant to the present study is that "language learning consists of the learning of constructions" and therefore "language teaching should consist of the teaching of constructions", and that "since constructions are form-meaning pairings, they should be presented as such" (Herbst, 2016, p. 41). He further suggests that teaching materials "be based on the analysis of corpora or on reference works based on corpus analysis and the frequency of constructions should be reflected in the design of teaching materials" (Herbst, 2016, p. 41). In a similar vein, Ellis et al. (2016: 272) noted that "[t]eaching materials should focus more on typical associations of lexical items and constructions and emphasize patterns in form-meaning relations". As shown in Fig. 2, the kind of constructicon developed in the present study not only adequately documents and describes the form-meaning-function pairings, but also offers quantitative information of each construction and each verb occurring in a given construction, thereby providing information that is fundamental for designing and developing academic writing teaching materials.

Overall, our study corroborates with the argument that language teaching and learning should "acknowledge the pervasiveness of constructions more than it currently does" (Römer et al., 2014: 967), which in turn points to the need to find ways in which constructions can be systematically identified and described. In the present study we propose that the integration of pattern grammar, local grammar and construction grammar represents one possible way to do so. Our study offers a nuanced description of the **V that** pattern, its associated local grammar patterns and the constructions, which not only has theoretical implications for construction grammar research but, more notably, pedagogical applications for academic writing teaching, as discussed above. The present study, then, demonstrates the feasibility and value of our proposed approach for EAP writing research and pedagogy.

## 6. Conclusion

The study has explored the possibility and feasibility of integrating pattern grammar, local grammar and construction grammar into EAP writing research and pedagogy. It proposes that local grammar analyses of grammar patterns could facilitate the identification of constructions and the subsequent development of constructicon, which in turn has important theoretical implications for construction grammar research and, more notably, pedagogical applications for academic writing teaching. In particular we noted that the construction network developed in the present study offers an inventory of form-meaning-function pairings available to express meanings, which can be incorporated into the development of academic writing teaching materials. We also pointed out that the construction network provides quantitative information about verbs occurring in each construction, which helps to raise learners' awareness of the association between verbs and constructions. These would indicate that our proposed approach can be regarded as an alternative methodology to EAP writing research and pedagogy.

The study reported in this paper can be extended in many ways. For instance, the study has only analysed the **V that** pattern and left other frequently occurring patterns (e.g., **N that**, ***it* v-link ADJ that/to-inf.**) in academic discourse unexplored. One direction for future research is thus to perform local grammar analyses of other highly frequent patterns and to systematise the constructions that can be identified thereby. Moreover, the study focused on the discipline of Linguistics. As studies have shown that language use varies according to disciplines (e.g., Charles, 2006a; Hyland, 2002), another important direction for future research is to extend the proposed approach to explore **V that** (and other frequent patterns) in other disciplines and, furthermore, disciplinary variations with regard to the uses of patterns and constructions. This would be of great value for disciplinary academic writing teaching. Third, future study can employ the proposed approach to explore learner writing and compare learners' use of patterns and constructions with that of expert writers', so as to diagnose the problem or difficulty learners may encounter in academic writing practice. Finally, it would also be worthwhile to explore how findings of the kind of research reported in this paper can be incorporated into the development of EAP

teaching materials and the effectiveness of such materials in EAP teaching practice. Arguably, all these lines of research can significantly contribute to EAP writing research and pedagogy.

## Statement

The study reported in this paper has not been published previously or submitted for consideration in any other journals.

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## CRediT authorship contribution statement

**Hang Su:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization. **Jun Ye:** Writing – review & editing, Writing – original draft, Visualization, Data curation. **Naixing Wei:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

## Declaration of competing interest

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**Hang Su** is a professor of Foreign Languages and Applied Linguistics at Sichuan International Studies University, China. His research interests mainly are corpus linguistics, (corpus) pragmatics, and L2/academic writing. He has published widely in *Applied Linguistics*, *ELT Journal*, *Journal of English for Academic Purposes*, *English for Specific Purposes*, *Assessing Writing*, among many others.

**Jun Ye** is currently a PhD candidate at Sichuan International Studies University, China. Her research interests include corpus linguistics, discourse analysis, and L2/academic writing. Her recent publication appeared in *Journal of Pragmatics* and *Corpus-based Studies across Humanities*.

**Naixing Wei** is a distinguished professor of Foreign Languages and Applied Linguistics at Beihang University, China. His research focuses on corpus linguistics, applied linguistics, and L2/academic writing. He has published widely in *International Journal of Corpus Linguistics, Corpus Linguistics and Linguistic Theory, Pragmatics* and many others.